

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 2 Mathematics

Assessment 1

March 2013

TIME ALLOWED: 70 minutes

Instructions:

- ***Start each question on a new page.***
- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- It is suggested that you spend no more than 5 minutes on Part A.
- Approved calculators may be used.

PART A: (5 Marks)

Answers to these multiple choice should be completed on the multiple choice answer sheet supplied with your answer booklet.

All questions are worth 1 mark

(a)	The value of i^{2014} is A. 1 B. -1 C. i D. $-i$
(b)	As the eccentricity of a standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ approaches zero, i.e., as $e \rightarrow 0$, what happens to the ellipse? A. It becomes a point B. It becomes a hyperbola C. It becomes more elliptical D. It becomes a circle.
(c)	The Cartesian form of the conic given by $x = 4\sec\theta$ and $y = 3\tan\theta$ is A. $\frac{x^2}{16} - \frac{y^2}{9} = 1$ B. $\frac{x^2}{9} - \frac{y^2}{16} = 1$ C. $\frac{x^2}{16} + \frac{y^2}{9} = 1$ D. $\frac{x^2}{9} + \frac{y^2}{16} = 1$
(d)	The length of the major axis of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is: A. 2 B. 3 C. 4 D. 6
(e)	What is the solution to the equation $z^2 = i\bar{z}$? (A) (0,0) and (0,1) (B) (0,0) and (0,-1) (C) (0,0), (0,-1), $(\frac{\sqrt{3}}{2}, \frac{1}{2})$ and $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$ (D) (0,0), (0,1), $(\frac{\sqrt{3}}{2}, \frac{1}{2})$ and $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$

PART B

(START EACH QUESTION ON A NEW PAGE)

QUESTION 1: (15 Marks)

Marks

- 3 (a) Let $x = 5 - i$ and $y = 3 + 4i$.
- Find (i) $|y|$ (ii) $\bar{x}$ (iii) $\frac{y}{x}$ (give your answer in the form $a + ib$)
- 1 (b) On separate Argand Diagrams, sketch the solutions to:
- 1 (i) $|z - 1| < 2$
- 1 (ii) $\frac{\pi}{4} < \arg(z - 1) < \frac{\pi}{3}$
- 1 (c) (i) If a point P on the hyperbola $xy = c^2$ has its x -value as $x = ct$, give its y -value
- 1 (ii) Find the equation of the tangent at P
- 2 (iii) If this tangent cuts the co-ordinate axes at A and B, show that $PA = PB$.
- 2 (d) If $z = 1 + \sqrt{3}i$, find
- (i) $\arg z$ (ii) z^6 , in simplest form
- 4 (e) If $|z| = 1$ and $\arg z = \theta$, show that $\arg\left[\frac{(z+1)^2}{z}\right] = 0$

QUESTION 2: (15 Marks) (Start on a new page)

Marks

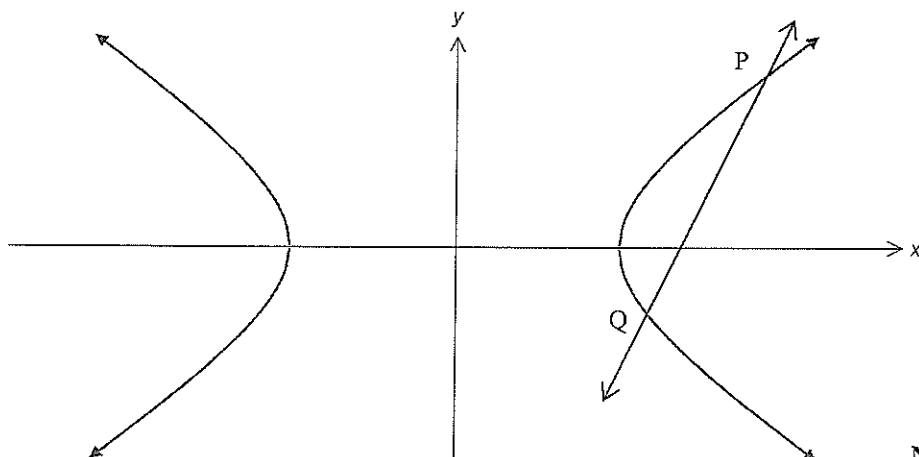
- 3 (a) Find the gradient of the tangent to the curve $x^4 + y^4 - 5xy^2 = 0$ at the point where $x=2$ and $y = \sqrt{2}$

- 1 (b) (i) Find the argument and modulus of $1 - i$

- 2 (ii) Hence, by using De Moivre's Theorem, or otherwise, simplify the expression

$$(1 - i)^8 + (1 + i)^8$$

- (c) P $(4\sec\theta, 3\tan\theta)$ and Q $(4\sec\alpha, 3\tan\alpha)$ are points on the Hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ with parameters θ and α , where $\theta + \alpha = \frac{\pi}{2}$ and $\alpha \neq \frac{\pi}{4}$.



NOT TO SCALE

- 2 (i) Find the co-ordinates of Q in terms of θ , in simplest trigonometric form.

- 2 (ii) Prove that the gradient of the chord PQ is $\frac{3}{4}(\cos \theta + \sin \theta)$

- 3 (iii) Find the equation of the chord PQ, in gradient/intercept form, and hence find the coordinates of a point on PQ that is independent of the value of θ .

- 2 (iv) As $\theta \rightarrow \frac{\pi}{2}$, show that the chord PQ approaches a line parallel to an asymptote of the hyperbola.

QUESTION 3: (15 Marks) (Start on a new page)

Marks

(a) For the ellipse $\frac{x^2}{4} + y^2 = 1$,

1 (i) Find the eccentricity, e .

2 (ii) Find an expression for $\frac{dy}{dx}$, and hence find the slope of the tangent at $P(x_0, y_0)$

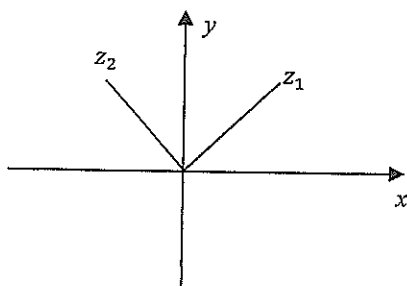
1 (iii) Prove that the equation of the tangent at P is $\frac{xx_0}{4} + y y_0 = 1$

1 (iv) The tangent at P meets the Directrix cutting the positive x-axis at Q.
Prove that the y-value of Q is $y_Q = \frac{\sqrt{3} - x_0}{\sqrt{3}y_0}$

1 (v) If $x_0 > 0$, and $y_0 > 0$, find the range of values of x_0 , so that Q lies below the x-axis.

(b) z_1 and z_2 , shown on the Argand Diagram below, are complex numbers such that

$$\frac{z_1 + z_2}{z_1 - z_2} = 2i,$$



(i) Copy the diagram onto your answer sheet (NO MARKS)

2 (ii) On the diagram, plot the points $z_1 + z_2$ and $z_1 - z_2$

2 (iii) Show that $|z_1| = |z_2|$

5 (c) The sequence $1, \sqrt{3}, \sqrt{1 + 2\sqrt{3}}, \dots$

has its n th position given by $x_n = \sqrt{1 + 2x_{n-1}}$

By the process of Mathematical Induction, prove that $x_n < 4$ for all $n \geq 1$

Multiple Choice Answer Sheet

Name _____

Completely fill the response oval representing the most correct answer.

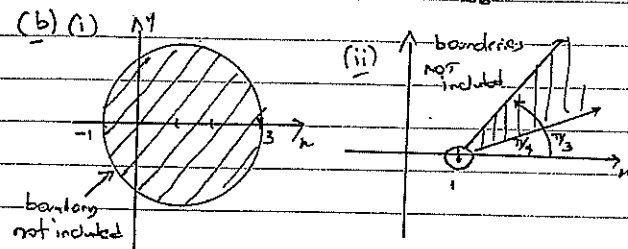
1. A ☐ B ☒ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☒
3. A ☒ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☐ C ☒ D ☐

PART B

Question 1:

(a) (i) 5 (ii) $5+i$ (iii) $\frac{11+23i}{26}$

1 MARK each



1 MARK EACH

(c) (i) $y = \frac{t^2}{x}$
 $= \frac{9}{t}$

1 MARK

(ii) $t^2 y + x = 2ct$ (or similar)

(iii)

At A, $x=0$

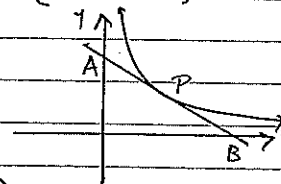
$\therefore A$ is $(0, 2ct)$

At B, $y=0$

$\therefore B$ is $(2ct, 0)$

MIDPOINT of AB is (ct, ct) which is P

$\therefore PA = PB$



1 MARK ONLY (working NOT required).

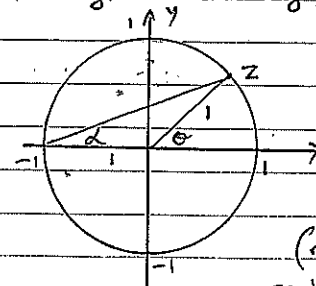
1 MARK

1 MARK

(d) (i) $\arg z = \pi/3$ (ii) $z = 2 \cos \pi/3$
 $z^6 = 64 \cos 2\pi$
 $= 64$

1 MARK each

(e) $|z|=1$ and $\arg z = 0$



From the diagram,

$\theta = 2\alpha$

(method I: external angle of a triangle)

(method II: angle at the centre is twice that at the circumference)

$\arg \left[\frac{(z+1)^2}{z} \right] = \arg (z+1)^2 - \arg z$
 $= 2\arg(z+1) - \arg z$
 $= 2\alpha - \theta$

2 for arriving here

2 for simplification

Question 2

(a) $4x^3 + 4y^3 \frac{dy}{dx} - 5(y^2 + x^2 y \frac{dy}{dx}) = 0$

1 MARK

$$\frac{dy}{dx} (4y^3 - 10xy) = 5y^2 - 4x^3$$

$$\frac{dy}{dx} = \frac{5y^2 - 4x^3}{4y^3 - 10xy}$$

1 MARK

At (2, √2)

$$m_T = \frac{10 - 32}{8\sqrt{2} - 20\sqrt{2}}$$

$$= \frac{1/6\sqrt{2}}{1/6\sqrt{2}} \text{ or } \frac{11\sqrt{2}}{12}$$

1 MARK

(b) (i) $\arg(1-i) = -\pi/4$ $|1-i| = \sqrt{2}$

1 MARK for EACH (n=2)

(ii) $(1-i)^8 + (1+i)^8 = [\sqrt{2}\cos(-\pi/4)]^8 + [\sqrt{2}\cos(\pi/4)]^8$

$$= 16\cos(-2\pi) + 16\cos(2\pi)$$

$$= 32$$

1 MARK

(c) (i) x-coordinate = $4\sec(\theta)$

$$= 4\sec(\pi/2 - \theta)$$

$$= 4\csc\theta$$

y-value is $3\cot\theta$

1 for each
= 2

(ii) $m_{PQ} = 3\cot\theta - 3\tan\theta$

$$= 4\csc\theta - 4\sec\theta$$

$$= \frac{3}{4} \left[\frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} \right] \left[\frac{1}{\sin\theta} - \frac{1}{\cos\theta} \right]$$

$$= \frac{3}{4} \left(\frac{\cos^2\theta - \sin^2\theta}{\cos\theta - \sin\theta} \right)$$

$$= \frac{3}{4} (\cos\theta + \sin\theta)$$

(iii) Equation of chord PQ is

$$y - 3\tan\theta = \frac{3}{4} (\cos\theta + \sin\theta) (x - 4\sec\theta)$$

$$4y - 12\tan\theta = 3(\cos\theta + \sin\theta)(x - 4\sec\theta)$$

$$4y = 3x(\cos\theta + \sin\theta) - 12$$

This goes through (0, -3)

→ (2) to get to here

(for y =)

1 MARK

(iv) $m_{PQ} = \frac{3}{4} (\cos\theta + \sin\theta)$

As $\theta \rightarrow \pi/2$ $m_{PQ} \rightarrow \frac{3}{4}$

← 1 MARK

as the asymptotes are $y = \pm \frac{3x}{4}$

1 for reasons

PQ is parallel to $y = \frac{3x}{4}$

the slope of
 $y = \frac{3x}{4}$

Question 3:

(a) (i) $b^2 = a^2(1-e^2)$

$$e^2 = -1/4 + 1$$

$$e = \sqrt{3}/2$$

1 MARK

(ii) $2x/4 + 2y/4 = 0$

$$\therefore \frac{dy}{dx} = -\frac{1}{2} \times \frac{1}{2y}$$

$$= -\frac{1}{4y}$$

$$m_T = -\frac{x_0}{4y_0}$$

1 MARK

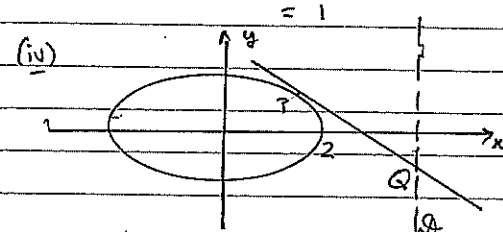
(iii) $y - y_0 = -\frac{x_0}{4y_0} (x - x_0)$

$$4yy_0 - 4y_0^2 = -xx_0 + x_0^2$$

$$x_0^2 + 4y_0^2 = xx_0 + 4yy_0$$

$$\therefore \frac{xx_0}{4} + yy_0 = \frac{x_0^2}{4} + y_0^2$$

1 MARK for
an effort
to prove.



Direction is $x = 4\sqrt{3}$

Q has y-value: $\frac{4\sqrt{3}x_0 + yy_0}{4} = 1$

$$\frac{x_0}{\sqrt{3}} + yy_0 = 1$$

$$y = \frac{1 - \frac{x_0}{\sqrt{3}}}{y_0}$$

$$= \frac{\sqrt{3} - x_0}{\sqrt{3}y_0}$$

1 MARK

(v) For $y_0 < 0$

$$\sqrt{3} - x_0 < 0$$

$$\therefore x_0 > \sqrt{3}$$

But $x_0 < 2$ (ellipse)

$$\sqrt{3} < x_0 < 2$$

2 MARKS

